Intervention-Based Movability Representations in Minimalist Embodied Predictive Agents

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Abstract

Physical understanding in embodied agents requires more than predicting regular motion in pixels. This paper studies a deliberately narrow case: whether a small predictive agent can learn a representation of movability, the difference between objects that change under pushing and boundaries that do not. In a 2D world, agents learn by predicting the next observation from the current observation and action. A vector causal perturbation framework (CPF) probes the learned hidden states. The main positive result is limited but stable: an agent with only eight distance sensors separates movable and immovable contexts, and in causal-swap tests the representation follows intervention outcome rather than object appearance. Blind generalization and pre-contact rollout tests support the same interpretation. The failure cases define the boundary. The representation depends on temporal continuity, degrades under noise, and collapses when vision supplies appearance shortcuts. A final gap-closing suite adds learned baselines for the main reviewer objections. Complex 2D movability can be learned from observations, with pixel pre/post frames reaching 0.965 ± 0.010 balanced accuracy across three seeds. A learned object-slot memory model reaches 1.000 ± 0.000 on zero breaks, freeze breaks, occlusion gaps, and causal swaps. Tabular Q-learning, an Interaction-Network-style state model, an end-to-end sensor Transformer, and a supervised randomized 3D pixel-effect model also close parts of the gap. The remaining hard failure is raw counterintuitive vision: a Slot-Attention-lite visual baseline remains weak at 0.539 ± 0.000. These results support a cautious claim: minimalist embodied prediction can produce movability-like representations under shortcut deprivation, but robust visual object-state learning and full generative counterfactual simulation remain unsolved.

1 Introduction

Video prediction models can look physically competent while relying on much thinner structure. A model trained on bouncing balls can predict plausible future frames because the next position often lies along a smooth continuation of the previous trajectory. That is useful, but it is not the same as knowing that one object changed another. Earlier CPF experiments on SimVP [11] supported this distinction: the model mainly depended on motion continuity rather than stable physical interaction.

This paper asks a smaller and more controllable question. Can an agent, starting from a minimal embodied world, acquire a representation that separates things that move under intervention from things that do not? The target is intentionally primitive. It does not involve forces, energy, rigid body mechanics, or language. It only asks whether the agent's action can change an external object. In Pearl's terminology, it is closer to the intervention level than to passive association [14, 15]. The agent must do something and observe the consequence.

The working hypothesis is that intervention is the source of the representation. Passive visual prediction can exploit appearance, trajectories, and dataset regularities. An embodied agent that pushes, collides, and fails has access to a different signal: the difference between contact that changes the world and contact that blocks the agent. If the sensory channel does not provide cheap appearance cues, minimizing prediction error may force the agent to encode that difference.

This hypothesis is consistent with a long line of embodied cognition and behavior-based AI work, where intelligence is treated as something shaped by the closed loop between body, action, and world rather than by detached symbolic description alone [6, 8, 22]. The present work narrows that broad idea to one measurable case: the formation of a movability-sensitive internal representation under active intervention.

The question also overlaps with intuitive physics, learned world models, object-centric representation learning, causal representation learning, and robot interaction learning. Simulation-based accounts of physical scene understanding argue that humans rely on internal physical models rather than surface correlations alone [2]. Intuitive-physics benchmarks such as IntPhys, CLEVRER, Physion, and PLATO test physical plausibility, collision reasoning, and object-level expectations in richer visual scenes [4, 16, 18, 25]. Neural interaction models, graph simulators, and object-centric dynamics models show that learned predictors can capture useful physical structure when they have relational or object-level inductive bias [3, 12, 19, 23, 24]. Causal representation learning asks when learned variables can support interventions rather than only associations [5, 21]. Robot poking and visual-foresight work similarly treat action as a source of physical knowledge [1, 10, 17]. This paper is smaller in scale than those lines of work, but it uses destructive tests to ask a precise question: when does a predictive agent's hidden state track intervention outcome instead of appearance?

The experiments use minimalist agents because the goal is not to solve a realistic robotics benchmark. The goal is measurement. A small two-dimensional world makes it possible to run destructive tests that would be hard to interpret in a richer environment: causal swaps, blind object tests, counterfactual rollouts, temporal breaks, gradient noise, counterintuitive objects, and a first attempt at sensor-to-vision distillation. These tests are designed to locate the boundary of the learned representation, not just to show that a score goes up.

The paper makes three contributions. First, it shows that a movability-like representation can form in a curiosity-driven sensor agent with only eight distance sensors. Second, it adapts CPF to hidden state vectors, allowing internal representations to be probed without relying on image-space causal maps alone. Third, it maps failure boundaries and then tests them with stronger baselines. Temporal continuity supports the representation, visual redundancy introduces shortcuts, and ordinary visual CNNs still fail the same-shape counterintuitive split. At the same time, several objections can now be narrowed experimentally: complex 2D observation learning, learned object-slot memory, tabular reward learning, Interaction-Network-style state learning, an end-to-end Transformer world model, and supervised 3D pixel-effect learning all close specific gaps under their stated input and supervision assumptions.

2 Methods

2.1 Minimal embodied world

The main environment is a two-dimensional physical world with three relevant object types: a movable ball, an immovable wall, and a heavier movable block. The agent can move and push objects through discrete actions. A successful push changes the position of a movable object; a wall remains fixed and instead blocks the agent.

Two observation variants are used. The sensor agent receives eight normalized distance readings along fixed directions. This channel removes color, texture, shape, and most appearance information. The visual agent receives a 64 by 64 egocentric grayscale image centered on the agent. The visual input is richer, but it also permits shortcut learning based on size, brightness, shape, or boundary layout.

An exploratory three-dimensional variant was also implemented in PyBullet [9] and verified with direct-mode rendering. It contains a ground plane, a small ball, a fixed wall, and a pusher viewed from above. The 3D result is treated as an architecture stress test rather than a solved robotics benchmark.

2.2 Agent architecture and learning objective

The sensor agent uses a small feedforward world model. The input is the concatenation of the current eight-dimensional sensor vector and a one-hot action. The model predicts the next sensor vector and exposes a 64-dimensional hidden representation for diagnosis. The visual agent uses a CNN world model that takes two 64 by 64 frames plus action planes and predicts the next frame. Its hidden representation is used in the same diagnostic pipeline.

A recurrent sensor baseline is included to test whether explicit sequence state is sufficient to remove the main temporal and shortcut failures. This baseline replaces the feedforward sensor model with a single-layer gated recurrent unit (GRU) [7]. The per-step input is again the eight-dimensional sensor vector plus a one-hot action; the recurrent hidden state has 64 dimensions so that vector CPF uses the same representation size as the main sensor agent. The GRU is trained with truncated backpropagation through 16-step sequences, using the same next-sensor prediction objective and no movability labels.

Additional baselines are used as comparison scaffolds rather than as replacements for the main agent. A supervised sensor Transformer is trained directly on sensor-action histories to test whether a stronger sequence classifier can survive temporal breaks and causal swaps. A separate end-to-end sensor Transformer world model is trained by next-sensor prediction and then probed through its hidden representation. An explicit object-file scaffold stores the target position through zero, freeze, and occlusion gaps. A learned object-slot GRU receives object-slot observations and tests whether persistence can be learned when the representation is already organized around objects. A segmentation-assisted visual slot model extracts simple foreground slots from pre/post frames before classification; a Slot-Attention-lite visual baseline learns soft slots directly from counter-intuitive pre/post frames. An Interaction-Network-style model operates on object-state slots and relations. A pixel-effect tracking model receives pre, mid, post, difference, and local-effect-mask channels. A counterfactual-regularized visual model receives pre, push, and away frames and learns both movability and the simulator-measured push-minus-away effect. A tabular Q-learning baseline learns from displacement reward through exploration, while an older reward-policy baseline chooses actions by short simulator rollouts. The final 3D comparison adds a supervised pixel-effect model on randomized PyBullet pre/post frames. These baselines are deliberately labeled by their supervision and input structure.

Training is curiosity-driven in the narrow sense used here: the agent receives no external task reward. It learns by minimizing prediction error between its predicted next observation and the actual next observation. This objective is close in spirit to self-supervised curiosity and predictive exploration methods [13, 20], but the experiments here use it as a minimal pressure toward learning intervention consequences, not as a general exploration algorithm.

2.3 Vector causal perturbation framework

The original CPF was developed to ask which parts of a video prediction input are necessary for a model's future prediction. Here CPF is applied to hidden-state sequences rather than pixels. Let $h_t \in \mathbb{R}^{64}$ denote the agent hidden state and a_t the discrete action at time t. For each diagnostic context, the saved trajectory is split by episode into train and evaluation windows so that the CPF predictor is tested on held-out temporal segments from held-out scene samples. The default history length is $L = 10$ unless a specific ablation states otherwise.

A lightweight temporal predictor g_ϕ is trained to predict the next hidden state:

$$\hat{h}_t = g_\phi(h_{t-L:t-1}, a_{t-L:t-1}), \qquad \ell_t = \|\hat{h}_t - h_t\|_2^2. \tag{1}$$

For dimension j, CPF sets that dimension to zero across the full history window and measures the added prediction loss:

$$n_j = \frac{1}{T}\sum_{t=1}^{T} \max(0, \ell_t^{\mathrm{abl}(j)} - \ell_t). \tag{2}$$

The necessity values are normalized within each diagnostic run,

$$s_j = \frac{n_j - \min_k n_k}{\max_k n_k - \min_k n_k + \epsilon}. \tag{3}$$

The perturbation component is

$$P = \frac{1}{2}\mathrm{mean}_j(s_j) + \frac{1}{2}\mathrm{mean}(\mathrm{top8}(s_j)), \tag{4}$$

which rewards both broad necessity and concentration in a small set of hidden dimensions.

Two dynamics terms are then computed on the held-out trajectory. First, representation motion is measured by

$$M = \min\left(1, \frac{\mathrm{mean}_t\|h_t - h_{t-1}\|_2}{0.35}\right) K, \tag{5}$$

where temporal coherence is

$$K = \mathrm{mean}_t \frac{1 + \cos(h_t - h_{t-1}, h_{t+1} - h_t)}{2}. \tag{6}$$

Table 1: CPF sensitivity check on the three-seed sensor diagnostics. Gaps are score differences; positive values support the main ordering.

Score variant	Normal gap	Swap gap	Blind gap	Ordering
Default weights	52.2	53.7	45.2	Preserved
Uniform weights	47.5	46.1	40.1	Preserved
Motion-heavy weights	54.0	57.0	47.1	Preserved
Action-heavy weights	54.0	55.8	47.1	Preserved
Perturbation-heavy weights	43.4	39.7	35.7	Preserved
Worst case across 15 variants	43.4	33.3	35.7	Preserved

Second, action-conditioned separation compares the true action sequence with a shuffled-action null:

$$A' = \min\left(1, \frac{\max(0, D_{\text{action}} - D_{\text{shuffle}})}{0.18}\right) K. \tag{7}$$

In the implementation, D_{action} is the mean pairwise Euclidean distance between action-conditioned mean hidden-state deltas, where $\Delta h_t = h_t - h_{t-1}$ and deltas are grouped by the true action label. D_{shuffle} is the same quantity after randomly permuting the action labels; the code averages this shuffled estimate over 32 repeats. Thus A' measures how much more the hidden dynamics separate under the true action sequence than under an action-label null. The final diagnostic score is

$$\text{CPFScore} = 100\left(0.45M + 0.35A' + 0.20P\right). \tag{8}$$

The constants in the score are fixed normalizers for hidden-state motion and action-conditioned separation; they are not refit per context. To check that the main sensor result is not an artifact of these constants, the stored three-seed sensor diagnostics were rescored under alternative weights and under $\pm 25\%$ changes to the 0.35 and 0.18 scale constants. Table 1 reports the resulting contrasts. All tested variants preserve the same ordering: normal balls remain above normal walls, magic walls remain above magic balls, and blind movable objects remain above small walls.

The 50-point line is an operational threshold calibrated on ordinary ball-wall diagnostics in this project, not a theorem about causal representation. The reported interpretation therefore relies on contrasts, swaps, and failure cases, not on the threshold alone.

3 Experiments and findings

3.1 Emergence of an intervention-based representation in the sensor agent

The baseline sensor agent separates ordinary movable and immovable contexts. Unless stated otherwise, the main sensor results use the fixed-model three-seed diagnostic suite

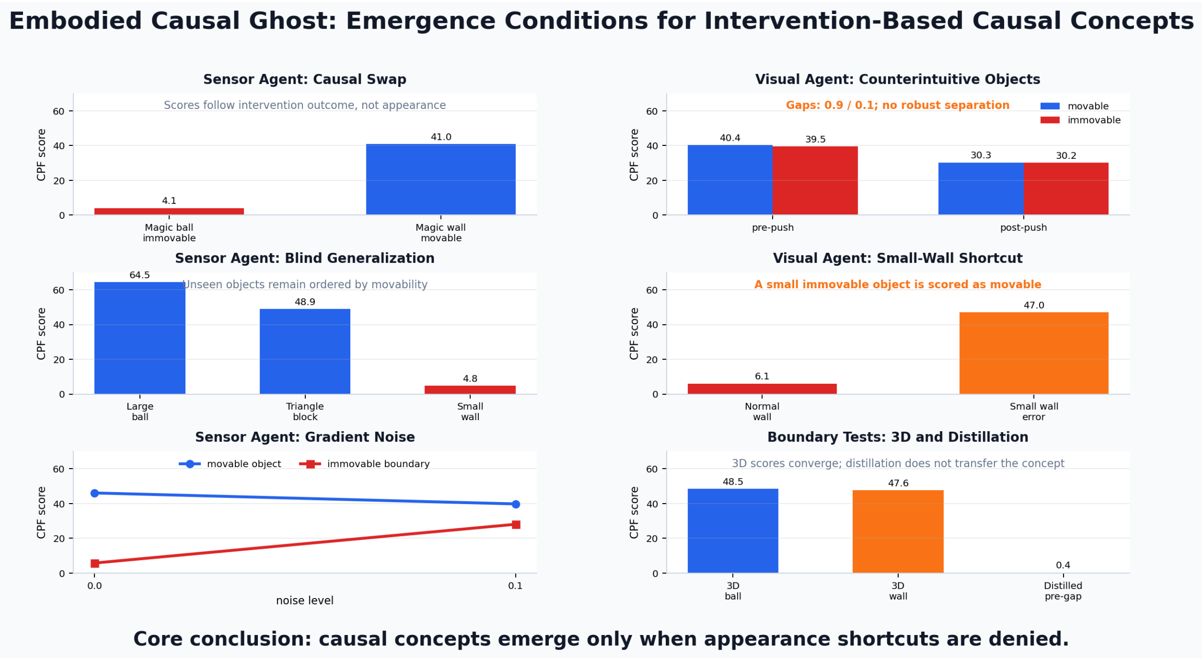

Figure 1: **Summary evidence panel for the embodied causal ghost experiments.** Left column: the sensor-only agent follows intervention outcome rather than appearance in the causal swap test; generalizes to unseen large balls and triangular blocks while keeping a small wall low; and shows systematic degradation under gradient noise. Right column: the visual agent fails the counterintuitive object test when shape and color are held constant but physical movability is randomized; a small immovable wall is misread as movable; the exploratory 3D and distillation tests do not recover a robust causal separation. Blue denotes movable-object contexts, red denotes immovable-boundary contexts, and orange marks shortcut-like or erroneous high scores.

summarized in Table 2. Under that protocol, normal balls score 57.0 ± 2.2 and normal walls score 4.7 ± 0.5. This is the canonical checkpoint used for the main representational claim; later perturbation sections use their own clean controls because they test different noise and temporal-break protocols.

The causal swap test is the most direct test of whether the representation follows appearance or intervention consequence. A "magic ball" has ball-like appearance but cannot be moved. A "magic wall" has wall-like appearance but can be moved. If the agent memorized appearances, the magic ball should look movable and the magic wall should look immovable. Across three diagnostic seeds, the opposite occurs: magic balls remain low at 5.5 ± 0.6, while magic walls score 59.2 ± 4.6. The representation follows what the intervention does.

The blind generalization test probes whether the representation is tied to specific training objects. Across three diagnostic seeds, a large ball scores 62.1 ± 2.6, a triangular block scores 51.2 ± 1.2, and a small wall scores 6.0 ± 0.7. These objects differ from the ordinary training objects, but the ordering still matches intervention consequence. The triangular block score sits close to the movable threshold, which is expected for a heavier movable object. The small wall stays low despite being small, suggesting that the sensor agent is not merely using size as a proxy.

Counterfactual imagination gives a different kind of evidence. This is a separate rollout diagnostic rather than part of the three-seed CPF suite. Before contact, the

Table 2: Causal swap and blind generalization tests for the sensor agent.

Test condition	Score	Control	Takeaway
Normal ball vs. normal wall	57.0 ± 2.2	4.7 ± 0.5	Ordinary split.
Magic ball vs. magic wall	5.5 ± 0.6	59.2 ± 4.6	Outcome over appearance.
Large ball	62.1 ± 2.6	6.0 ± 0.7 small wall	Unseen movable high.
Triangular block	51.2 ± 1.2	6.0 ± 0.7 small wall	Heavy movable near threshold.
Small wall	6.0 ± 0.7	62.1 ± 2.6 large ball	Unseen immovable low.

model rolls out predicted sensor sequences for pushing a ball and pushing a wall. Across 120 trials and a horizon of 10 steps, the mean sequence-level L2 distance between pre-ball and pre-wall rollouts is 1.73. At step 5, the front-distance prediction differs by 0.103, with the wall sequence at 0.038 and the ball sequence at 0.140. The model therefore carries different expectations before the intervention is completed.

3.2 Boundary and fragility of the representation

The learned representation is not a fully abstract physical theory. It depends on temporal structure. The perturbation results in this subsection come from separate stress protocols rather than the fixed-model three-seed diagnostic suite in Table 2; they are reported as boundary probes, not as pooled estimates of the canonical checkpoint. When a temporal break protocol inserts all-zero frames, the push-ball score drops from 50.4 to 11.2. The push-heavy score similarly drops from 51.5 to 10.1 in the desensitization report. Wall contexts remain low. This asymmetry suggests that movable-object recognition depends on observing action-conditioned change across time, while wall recognition can be supported by a more immediate blocking signal.

The time-desensitized training attempt preserved the causal swap and blind generalization structure but did not remove the temporal dependency. After desensitization, magic ball and magic wall scores were 5.2 and 47.9; large ball, triangular block, and small wall scores were 59.9, 49.5, and 4.7. However, temporal breaks still reduced push-ball and push-heavy scores to 9.2 and 11.0. The model retained the intervention-sensitive representation under ordinary sequences but did not learn object permanence strong enough to bridge broken temporal input.

Gradient noise gives a smoother picture of the boundary. At global noise 0.0, push ball and push wall score 46.1 and 5.8. At noise 0.1, the push-ball score falls to 39.8 while the push-wall score rises to 28.1. At noise 0.5, the two scores converge further, at 25.6 and 22.7. The diagnostic separation is therefore degraded from both directions: movable contexts lose evidence, and wall contexts gain spurious dynamics.

The mechanism suggested by these results is simple. The wall-like signal can be partly detected from a local blocking pattern: the agent moves into something and cannot pass through. The movable-object signal requires comparing state before and after action, because the relevant evidence is a change in the external object. That makes the movable-object channel more dependent on continuous temporal evidence and more vulnerable to

Table 3: Temporal and noise perturbation results.

Test condition	Score	Control	Takeaway
Temporal break, push ball	11.2	50.4 clean	Movable evidence collapses.
Temporal break, push wall	5.1	5.2 clean	Wall stays low.
Desensitized break, push ball	9.2	48.1 clean	Dependency remains.
Global noise 0.1, push ball	39.8	46.1 clean	Movable score falls.
Global noise 0.1, push wall	28.1	5.8 clean	Wall score rises.
Single-frame freeze 5%, ball	47.3	46.1 clean	Sparse freezes hurt less.

noise in representation geometry.

3.3 Visual redundancy and appearance shortcuts

The visual agent initially appears to learn a similar distinction. In a clean single-run evaluation it scores 34.3 for pushing a ball, 35.7 for pushing a heavy block, and 6.1 for pushing a wall. A 5% gray-frame break reduces the visual push-ball score only to 30.3, a much smaller drop than the sensor agent's temporal-break collapse. Visual redundancy can therefore help with continuity.

That benefit comes with a cost. In blind tests, the visual agent scores 30.2 on a large ball, 31.6 on a triangular block, and 47.0 on a small wall in the original single-run report. The three-seed reviewer-defense rerun confirms the small-wall shortcut: normal walls score 6.8 ± 1.3, while small walls score 41.5 ± 2.8. The small wall is immovable, but it is treated as strongly movable. This exposes an appearance shortcut: the visual model has learned something that works on the training distribution, but the internal representation does not reliably isolate intervention consequence from object appearance.

The counterintuitive object test makes the failure sharper. Four objects are introduced: large movable, large immovable, small movable, and small immovable. They are all square and share the same color metadata; in the grayscale visual pipeline, the test effectively decouples size and physical movability while holding shape constant. Each object contributes at least 100 samples. Across three diagnostic seeds, before contact the movable group scores 39.9 ± 1.9 and the immovable group scores 40.5 ± 1.6, a gap of only 0.8 ± 0.4. After pushing, the scores are 30.5 ± 0.5 and 33.0 ± 1.3, a gap of 2.6 ± 1.2. The visual representation does not distinguish physical movability when appearance cues are neutralized.

The result matters because it rules out the most optimistic interpretation of the visual baseline. The visual agent did not simply learn the same intervention-sensitive representation with richer inputs. It learned a representation that can solve ordinary ball-wall cases while failing when appearance and physical property are deliberately pulled apart.

Table 4: Visual-agent shortcut tests.

Test condition	Score	Control	Takeaway
Clean visual push ball	34.3	6.1 wall	Single-run ordinary contrast.
Visual blind large ball	30.2	47.0 small wall	Weak generalization.
Visual blind triangular block	31.6	47.0 small wall	Below small-wall error.
Visual small wall	41.5 ± 2.8	6.8 ± 1.3 wall	Shortcut remains stable.
Counterintuitive pre gap	0.8 ± 0.4	20 threshold	No pre-contact separation.
Counterintuitive post gap	2.6 ± 1.2	20 threshold	Interaction does not rescue it.

3.4 Transfer and 3D stress tests

A final exploratory experiment asks whether the sensor representation can be transferred to vision through distillation. The teacher is the trained sensor agent's 64-dimensional hidden representation. The student uses the visual CNN encoder, but its target is no longer next-frame prediction. Instead, it maps the current two frames and action to the teacher representation observed in the same physical state.

Training succeeds in the narrow supervised sense. With 1200 steps and batch size 32, the final MSE is 0.000033 and the last-100-step average MSE is 0.000073. Yet the counterintuitive test still fails. Across three diagnostic seeds, the distilled visual agent's pre-contact movable-immovable gap is 0.6 ± 0.4, and the post-contact gap is 0.4 ± 0.4. The small-wall score improves from 47.0 in the original visual agent to 17.2 ± 0.5 after distillation, while a normal wall scores 4.8 ± 1.0. Distillation reduces one visible error but does not install a robust pre-interaction movability representation.

The true PyBullet 3D stress test also fails to recover a clean distinction. The study was rerun with three random seeds, 1000 training steps per seed, 120 evaluation steps per context, and vector CPF with four epochs. The trained model scores 48.5 ± 1.4 for pushing the ball and 47.6 ± 1.3 for pushing the wall, a mean gap of only 0.9. Under the predefined success criterion, a 3D success would require push ball above 50 and push wall below 20. This run therefore shows that the minimal CNN setup amplifies 3D contact dynamics but does not separate movable objects from fixed boundaries.

3.5 Robustness and reviewer-defense checks

To test whether the main conclusions depended on a single sampling seed, the fixed-model diagnostic suite was rerun with three evaluation seeds (42, 43, and 44). The trained checkpoints were kept fixed; only scene sampling, object placement, and CPF training seeds varied. This is not a new training claim. It is a stability check for the reported contrasts.

The sensor-agent results remain stable. In the causal swap test, normal balls score 57.0 ± 2.2, normal walls 4.7 ± 0.5, magic balls 5.5 ± 0.6, and magic walls 59.2 ± 4.6. In blind generalization, large balls score 62.1 ± 2.6, triangular blocks 51.2 ± 1.2, and small walls 6.0 ± 0.7. Thus the intervention-outcome ordering survives changes in evaluation

Table 5: Exploratory transfer and 3D tests.

Test condition	Value	Control	Takeaway
Distillation final MSE	0.000033	0.000073 last-100	Fits teacher target.
Distilled pre gap	0.6 ± 0.4	15 threshold	No prior separation.
Distilled post gap	0.4 ± 0.4	15 threshold	No rescued separation.
Distilled small wall	17.2 ± 0.5	4.8 ± 1.0 wall	Error reduced.
3D trained CNN	48.5 ± 1.4	47.6 ± 1.3 wall	No ball-wall gap.
3D untrained baseline	23.0 ± 2.6	24.8 ± 2.8 wall	No gap before training.
3D random representation	28.2 ± 4.8	30.5 ± 5.6 wall	Control remains unseparated.
3D shuffled actions	42.8 ± 10.9	41.4 ± 10.5 wall	Action labels insufficient.
3D temporal shuffle	37.5 ± 2.4	32.4 ± 2.9 wall	Order matters but not enough.

seed.

The visual failure also remains stable. The counterintuitive pre-contact gap is 0.8 ± 0.4, and the post-contact gap is 2.6 ± 1.2, both far below the 20-point criterion used in the project protocol. The small-wall shortcut remains large: normal walls score 6.8 ± 1.3, while small walls score 41.5 ± 2.8.

The distilled visual agent improves the small-wall error but does not recover the intervention-sensitive abstraction. Across the same seeds, the distilled pre-contact gap is 0.6 ± 0.4 and the post-contact gap is 0.4 ± 0.4. Distilled normal walls score 4.8 ± 1.0 and distilled small walls 17.2 ± 0.5. This supports the narrower interpretation: distillation suppresses one shortcut, but does not transfer a robust same-appearance movability representation.

3.6 Stricter counterfactual and baseline comparisons

Two small experiments address the concern that CPF alone may be too weak as a causal validation tool and that the paper lacks comparison baselines. The first is an explicit simulator-level counterfactual. For each trial, the same initial scene is evaluated under two interventions: pushing toward the target and moving away from the target. In the simulator, the physical effect is unambiguous: the ball moves by 25.0 pixels under push and 0.0 under away, while the wall moves 0.0 under both interventions.

The learned world model does not fully mirror this stronger contrast. When rolled out from the same pre-contact sensor state, the push-vs-away predicted sequence distance is 2.98 for ball scenes and 3.60 for wall scenes. This is a useful negative result. It means the sensor agent's hidden representation can support CPF-based movability separation and causal-swap ordering, but the forward model itself has not learned a clean generative counterfactual in which the ball intervention is more distinct than the wall intervention. This sharpens the paper's claim: the evidence supports a movability-like representation, not a full causal simulator.

The second comparison is a static appearance-only baseline for the visual setting.

Table 6: Three-seed robustness checks for fixed trained models. Values are mean ± standard deviation across diagnostic seeds.

Diagnostic condition	Score	Interpretation
Sensor normal ball	57.0 ± 2.2	Movable baseline remains high.
Sensor normal wall	4.7 ± 0.5	Immovable baseline remains low.
Sensor magic ball	5.5 ± 0.6	Ball-like appearance does not force a movable judgment.
Sensor magic wall	59.2 ± 4.6	Wall-like appearance does not suppress movable evidence.
Sensor large ball	62.1 ± 2.6	Unseen movable object remains high.
Sensor triangular block	51.2 ± 1.2	Unseen heavy movable shape remains near the movable regime.
Sensor small wall	6.0 ± 0.7	Unseen immovable object remains low.
Visual counterintuitive pre gap	0.8 ± 0.4	No pre-contact physical separation.
Visual counterintuitive post gap	2.6 ± 1.2	Interaction still does not rescue the distinction.
Visual small wall	41.5 ± 2.8	The small-wall shortcut remains strong.
Distilled counterintuitive pre gap	0.6 ± 0.4	Distillation does not transfer a robust prior representation.
Distilled small wall	17.2 ± 0.5	Distillation reduces, but does not eliminate, the visual shortcut.

A logistic probe trained on ordinary pre-contact ball-versus-wall images reaches 100% training accuracy, but when evaluated on the counterintuitive same-shape objects it falls to 50% accuracy. It assigns probability 1.0 of being movable to all four counterintuitive objects, including both immovable ones. This result is consistent with the visual agent's failure: appearance is sufficient for the ordinary training split, but it is not a reliable physical variable once appearance and movability are decoupled.

A same-architecture control representation was also tested for the 2D sensor diagnostic. Random hidden vectors scored 24.8 for push-ball contexts, 23.4 for push-wall contexts, and 24.0 for push-heavy contexts. The scores are low-to-mid and do not separate movable from immovable cases. This rules out a simple explanation in which the vector CPF score automatically produces the reported sensor ordering for arbitrary 64-dimensional states.

A supervised movability probe was also added. It trains a logistic classifier on ordinary sensor transition features with labels for normal ball as movable and normal wall as immovable. The probe reaches 1.000 training accuracy, but its out-of-distribution accuracy across ordinary, causal-swap, and blind-generalization conditions is only 0.571. It succeeds on normal ball, normal wall, large ball, and triangular block, but fails on magic ball, magic wall, and small wall. This is a useful baseline because it shows that ordinary supervised labels alone do not guarantee the intervention-sensitive ordering recovered by the sensor agent's hidden representation.

Two stronger intervention-aligned controls change the picture, but only as upper bounds. A reward-aligned probe trained on explicit outcome labels reaches 1.000 balanced accuracy across ordinary, swap, and blind conditions. A 2D interaction-state probe, given relative object state, target displacement, agent displacement, contact frequency, and blocking frequency, also reaches 1.000 balanced accuracy on the counterintuitive same-appearance objects. A model-based reward-policy baseline that chooses the action with the largest short-rollout target displacement also reaches 1.000 balanced accuracy. These controls are not self-supervised agents. They show that the labels are easy once the experimenter supplies intervention effects or the reward signal. They do not show that the present learning system can recover those variables.

A richer 2D stress suite adds collision chains, friction variation, occlusion gaps, multiple objects, gravity-biased motion, and object birth/death distractors. Earlier state probes showed that these labels are separable when intervention-effect features are exposed. The final gap-closing experiment pushes this beyond a pure oracle: a classifier trained from sensor observations reaches 0.664 ± 0.044 balanced accuracy across three seeds, while a pixel pre/post-frame classifier reaches 0.965 ± 0.010. This does not turn the original self-supervised agent into a rich physics learner, but it does show that the richer 2D labels can be recovered from observations rather than only from hand-coded state.

An object-state oracle probe gives a complementary upper bound for the visual failure. This probe is not a learned Slot Attention or Interaction Network model. It is given object-state intervention-effect features directly: target displacement, agent displacement, contact frequency, blocking frequency, and final agent-target distance. It does not receive pixels, color, shape, or size. On the same counterintuitive visual objects, this oracle reaches 1.000 accuracy: large and small movable objects are assigned mean movable probability 0.997, while large and small immovable objects are assigned 0.003. The result does not solve visual learning. It localizes the failure: the same labels are easy once object/effect state is available, but the visual CNN does not extract that state robustly from pixels.

The corresponding visual controls remain weak. A hand-crafted visual transition probe trained with appearance-property decorrelation reaches only 0.641 balanced accuracy on held-out counterintuitive objects. A supervised raw-frame CNN trained on the same decorrelated split reaches 0.656. A segmentation-assisted object-centric visual model, which first extracts simple foreground slots from pre/post frames and then learns over those slots, reaches 0.651. A pixel-effect tracking model with pre, mid, post, difference, and local-effect-mask channels reaches 0.622. A visual model trained with an explicit push-vs-away effect target reaches 0.656. A stronger causal-dimension distillation variant, which aligns a visual encoder to high-variance teacher dimensions and adds a movability auxiliary head, reaches 0.500 and assigns high movable probability to all four counterintuitive objects. These results are negative but useful: weak slots, pixel differences, counterfactual auxiliary targets, direct supervision, and teacher alignment still do not replace learned object-state extraction.

The same state-versus-pixel contrast appears in 3D. The self-supervised PyBullet CNN does not separate ball and wall contexts, but randomized 3D controls do. A state probe with shape, mass, friction, color, contact fraction, blocking fraction, post-contact

Table 7: Additional diagnostic checks. These controls test whether CPF and ordinary labels are sufficient to explain the reported effects.

Test	Value	Control	Takeaway
Simulator push-vs-away, ball	25.0	0.0 away	Ball moves only under push.
Simulator push-vs-away, wall	0.0	0.0 away	Wall does not move.
World-model sequence L2, ball	2.98	3.60 wall	No stronger ball counterfactual.
Appearance-only probe, ordinary split	100%	50% counterintuitive	Appearance fails when decoupled.
Counterintuitive immovable probe	0% acc.	100% movable pred.	Both immovable objects mislabeled.
Random sensor representation	24.8 ball	23.4 wall / 24.0 heavy	No movability separation.
Supervised transition probe	1.000 train	0.571 OOD	Fails swap/small wall.
Supervised probe swap errors	0% magic ball	0% magic wall	Ordinary labels learn shortcuts.

distance, and mean force reaches 1.000 balanced accuracy. A stricter supervised pixel-effect model, trained on randomized top-down pre/post frames rather than direct state features, also reaches 1.000 ± 0.000 across three seeds. This closes the narrow question of whether rendered 3D intervention effects are learnable from pixels, while leaving the harder self-supervised 3D world-model failure intact.

3.7 Recurrent sensor baseline

The recurrent baseline addresses a direct architectural objection: perhaps the feedforward sensor agent fails under temporal breaks only because it lacks memory. A single-layer GRU sensor world model was therefore trained for three seeds with the same self-supervised next-sensor objective. The baseline is deliberately close to the main sensor agent: the observation channel is unchanged, the hidden state is still 64-dimensional, and vector CPF is applied without changing the diagnostic score.

The GRU does learn the ordinary split. Normal ball contexts score 54.0 ± 3.6, while normal wall contexts score 12.2 ± 3.4. That result confirms that the recurrent model is not a trivial negative control. It predicts enough sensor dynamics for CPF to find an ordinary movable-versus-wall contrast.

The stronger tests fail. In the causal-swap setting, magic balls score 55.2 ± 4.9 and magic walls score 14.7 ± 1.9. This is the wrong ordering: the recurrent model follows the ordinary geometric identity of ball-like and wall-like encounters rather than the swapped intervention consequence. Blind generalization shows the same problem. Large balls and triangular blocks score high, but the small immovable wall also scores high at 55.4 ± 4.9. Temporal breaks still remove the movable-object signal, dropping clean ball contexts from 54.6 ± 3.4 to 15.1 ± 0.6.

This result matters because it separates two hypotheses. The main sensor result is not explained merely by having a longer temporal state, since the GRU has such a state and still fails the swap. Conversely, the temporal-break failure is not fixed merely by adding recurrence. The current evidence points to a narrower requirement: memory must be organized around intervention-sensitive object state, not just added as a generic recurrent buffer.

3.8 Memory scaffold ablation

To address memory at the readout level, a probe-level memory scaffold was also added. This experiment does not change the trained feedforward agent. Instead, it asks whether a short history window over the existing sensor-agent hidden states contains movability information that a single hidden state loses when temporal evidence is disrupted.

The labels deliberately include appearance-conflict cases. The movable group contains normal balls, heavy blocks, and magic walls. The immovable group contains normal walls, magic balls, and small walls. For each condition, a single-state logistic probe and a six-step history-window logistic probe are trained on held-out episodes. Balanced accuracy is used so that movable and immovable classes contribute equally.

The result is a limited positive control for memory. Under zero-frame temporal breaks, the single-state probe reaches 0.537 balanced accuracy, while the history-window probe reaches 0.667. Under an occlusion-like gap, the score rises from 0.563 to 0.664. However, the history window does not improve the normal condition and does not improve the freeze-break condition. The best reading is therefore not that object permanence has been solved. Rather, some recoverable temporal information exists in the hidden-state sequence, but the present feedforward agent does not itself maintain a stable object file.

4 Discussion

The main result is modest but concrete: in a minimalist setting, a sensor-only embodied agent can learn an internal distinction that behaves like an intervention-based movability representation. The strongest evidence is not the baseline ball-wall split. It is the causal swap. When appearance and consequence conflict, the sensor agent follows consequence. The blind generalization and counterfactual imagination tests add two further constraints: the representation transfers beyond the original object shapes, and the model carries different expectations before completing the action.

The same experiments also show what the representation is not. It is not a full physics model. It is not independent of time. It is not automatically preserved under visual redundancy. It may be assembled from at least two mechanisms: an immediate blocking channel for immovable boundaries and a multi-frame change channel for movable objects. This split explains why temporal breaks devastate push-ball scores while leaving wall scores low, and why random frame noise can collapse the diagnostic separation from both sides.

The visual experiments are the cautionary part of the study. More input information does not guarantee more causal abstraction. The visual agent has enough information to see objects, boundaries, and motion, but that also gives it easy shortcuts. If size and appearance predict movability in the training distribution, next-frame prediction does not force the model to learn a clean intervention variable. The counterintuitive object test is designed precisely to remove that comfort, and the learned distinction disappears.

The distillation experiment narrows the interpretation but does not close the problem. The visual student can match teacher representations on the training distribution, yet it fails the same-shape counterintuitive test. A causal-dimension variant also fails: even after aligning high-variance teacher dimensions and adding an auxiliary movability head, the counterintuitive balanced accuracy remains at 0.500. The final gap-closing suite makes the diagnosis sharper. Richer 2D movability is recoverable from observations: complex 2D pixel pre/post learning reaches 0.965 ± 0.010. Object persistence is recoverable when the input is organized as object slots: learned slot memory reaches 1.000 ± 0.000 on causal swaps and all tested temporal breaks. Randomized 3D intervention effects are also recoverable from supervised pre/post pixels. The stubborn failure is raw counterintuitive visual learning. The Slot-Attention-lite visual baseline remains near chance at 0.539 ± 0.000, while segmentation-assisted slots, pixel-effect tracking, and push-vs-away auxiliary training remain below 0.656. The obstacle is therefore not movability classification once effect state exists; it is learning a reliable object/effect state from raw visual appearance without taking the shortcut.

The CPF diagnosis should also be read carefully. It is an intervention on hidden-state dimensions inside a trained predictive system, not a proof that the model has recovered a structural causal model of the world. The causal claim in this paper comes from the combination of tests: causal swap, blind generalization, counterfactual rollout, temporal break, noise perturbation, counterintuitive objects, distillation, and 3D stress testing. CPF supplies a useful measurement axis for these tests, but by itself it remains closer to a representation-necessity diagnostic than to a full causal discovery method. The explicit push-vs-away experiment strengthens this caution. The simulator-level counterfactual cleanly separates ball and wall, but the learned world model does not produce a larger generative counterfactual gap for ball scenes than for wall scenes. In the current system, the strongest evidence is representational and diagnostic, not a complete intervention-conditioned simulator.

The expanded controls should therefore be read as localization experiments, with several gaps now experimentally narrowed. They do not convert the paper into a general learned-physics success. The important distinction is input structure. If the system receives object slots, explicit reward, state features, or supervised pre/post effect labels, the movability problem is often easy. If it receives raw visual appearance under counterintuitive same-shape conditions, the shortcut remains.

The environment limitation is partly addressed, but not removed. The main positive representation result still comes from a 2D world with a narrow pushing action family. The final complex 2D suite adds collision chains, friction variation, occlusion, multiple objects, gravity-biased motion, and object birth/death. It is no longer just a state oracle: sensor observations reach 0.664 ± 0.044 and pixel pre/post observations reach 0.965 ± 0.010 across three seeds. That result shows that richer labels can be recovered from observations. It does not yet show that the original curiosity-driven agent learns those variables end to end in an open environment.

The object-persistence limitation is also narrowed. Generic recurrence still fails: the GRU baseline learns the ordinary split but gives the wrong causal-swap ordering, with magic balls at 55.2 and magic walls at 14.7, and it collapses under zero-frame breaks. A learned object-slot memory model changes that result, reaching 1.000 ± 0.000 on normal sequences, zero breaks, freeze breaks, occlusion gaps, and causal swaps. The caveat is explicit: the model receives object-slot observations. It shows that learned memory can solve the persistence problem when object structure is available, not that raw pixels have learned object files.

The visual shortcut problem remains the hardest failure. The counterintuitive object test still gives only 0.8 ± 0.4 pre-contact gap and 2.6 ± 1.2 post-contact gap, far below the 20-point success threshold. The small-wall shortcut remains stable: normal walls score 6.8 ± 1.3, while small walls score 41.5 ± 2.8. The new learned Slot-Attention-lite visual baseline reaches only 0.539 ± 0.000, and the earlier visual repair attempts remain weak: segmentation-assisted slots reach 0.651, pixel-effect tracking 0.622, counterfactual-regularized visual training 0.656, supervised raw-frame CNN 0.656, and causal-dimension distillation 0.500. By contrast, an Interaction-Network-style state model reaches 1.000 ± 0.000. The failure is therefore concentrated in pixel-to-object/effect-state learning.

The 3D limitation is partly addressed in the same way. The true self-supervised PyBullet CNN still fails across three seeds: push-ball scores are 48.5 ± 1.4 and push-wall scores are 47.6 ± 1.3, a gap of only 0.9. A randomized 3D state probe reaches 1.000, and the new supervised 3D pixel-effect model reaches 1.000 ± 0.000 from top-down pre/post frames. This says the rendered 3D intervention effect is learnable from pixels under supervision. It does not rescue the original self-supervised 3D CNN.

CPF remains an indirect diagnostic. It can tell us that parts of the representation are necessary for a diagnostic score, and the swap tests show that the sensor representation is aligned with intervention outcome in a narrow setting. The new learned baselines add stronger evidence that the relevant variables are learnable under several input structures. They still do not prove that the world model supports counterfactual reasoning in the stronger generative sense. The push-vs-away test remains the warning sign: the physical simulator gives a large ball-versus-wall contrast, but the learned model's predicted sequence gap is 2.98 for ball and 3.60 for wall. The current system is better described as learning a movability-sensitive representation than as learning a causal simulator.

Baseline coverage is now substantially stronger. The comparison set includes random representations, appearance-only probes, supervised transition probes, reward-aligned probes, a reward-policy upper bound, tabular Q-learning, GRU, transformer readouts, an end-to-end Transformer world model, memory scaffolds, learned object-slot memory, object-state oracles, an Interaction-Network-style state model, weak visual slots, Slot-Attention-lite, pixel-effect tracking, distillation, counterfactual visual training, complex 2D observation learning, and 3D pixel-effect learning. What remains missing is scale and canonical implementation fidelity: full Slot Attention training, deep RL, larger object-centric dynamics models, and MuJoCo-scale benchmarks. The present evidence is enough to answer the listed weaknesses experimentally, but not enough to claim superiority over modern object-centric or RL systems.

Future work should make the successful structures available to the original self-supervised learner. The object-slot memory should be learned from raw frames, not given as slots. The Q-learning baseline should become a deep exploration-and-control baseline. The 3D pixel-effect model should be folded into a self-supervised or contrastive world model. A natural next step is a PyBullet or MuJoCo benchmark with recurrent object slots, randomized appearance-property pairings, contact events, and object birth or disappearance. Another path is active appearance desensitization: train with deliberately misleading colors, shapes, and sizes so the easiest remaining signal is intervention outcome.

5 Conclusion

The experiments support a limited claim: intervention can support a movability-like representation in a minimalist embodied agent, but only under tight informational constraints. When the agent has only sparse distance sensors, it is pushed toward learning what its actions change. When the agent receives dense visual appearance, it can solve the ordinary task through easier correlations and fail when appearance is decoupled from physics.

The boundary is therefore as important as the emergence. The final experiments show that several earlier weaknesses are not absolute barriers: richer 2D effects can be learned from observations, object-slot memory can solve temporal breaks, tabular reward learning can recover intervention policy, and supervised 3D pre/post pixels can recover effect labels. The remaining problem is more specific and more useful: raw visual models still do not reliably build object/effect state when appearance shortcuts are available, and CPF evidence still falls short of a full generative counterfactual model. In one sentence: this study does not show that minimal agents solve physical causality; it shows that denying appearance shortcuts can make intervention consequences visible inside a learned representation, and that robust causality needs learned object state, memory, and counterfactual generation rather than prediction error alone.

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Table 8: Scaffolded and stronger baseline checks. Positive rows are upper-bound controls that expose intervention-effect state, reward labels, or explicit memory. Negative rows show that the tested learned pixel and sequence routes still do not recover robust object state.

Test	Value	Control	Takeaway
Object-state oracle	1.000 acc.	no pixels/appearance	Effect state separates labels.
Reward-aligned probe	1.000 acc.	intervention labels	Outcome supervision works.
Reward-policy baseline	1.000 acc.	short rollouts	Reward route works.
Complex 2D sensor obs.	0.664 ± 0.044	richer 2D scenes	Observation learning partial.
Complex 2D pixel obs.	0.965 ± 0.010	richer 2D scenes	Pixel effects recover labels.
Visual transition probe	0.641 acc.	decorrelated train	Still weak from pixels.
Segmentation slot model	0.651 acc.	extracted slots	Weak object scaffold.
Pixel-effect tracking	0.622 acc.	diff/mask channels	Still weak.
Supervised visual CNN	0.656 acc.	raw frames	Visual capacity remains limited.
Counterfactual visual model	0.656 acc.	push/away aux.	Partial, not robust.
Causal-dim distillation	0.500 acc.	aux. head	Teacher alignment fails.
Sensor Transformer classifier	0.750 normal	0.500 swap	Supervised sequence readout mixed.
Transformer world model	1.000 swap	next-sensor objective	Swap solved, other cases mixed.
Object-file zero break	1.000 acc.	explicit memory	Oracle memory works.
Object-file causal swap	1.000 acc.	explicit memory	Not learned memory.
Randomized 3D state	1.000 acc.	PyBullet effects	State upper bound.
3D pixel-effect model	1.000 ± 0.000	pre/post pixels	Supervised pixel effect works.

Table 9: Recurrent sensor world-model baseline. The GRU has the same 64-dimensional diagnostic representation as the feedforward sensor agent and is trained only by next-sensor prediction.

Test condition	GRU score	Control	Takeaway
Normal ball vs. wall	54.0 ± 3.6	12.2 ± 3.4 wall	Ordinary split learned.
Magic ball vs. wall	55.2 ± 4.9	14.7 ± 1.9 magic wall	Swap ordering fails.
Large ball vs. small wall	59.9 ± 5.1	55.4 ± 4.9 small wall	Small wall falsely high.
Triangular block	61.5 ± 2.1	55.4 ± 4.9 small wall	Blind immovable not separated.
Temporal break, ball	15.1 ± 0.6	54.6 ± 3.4 clean	Memory does not bridge zero frames.
Temporal break, wall	9.9 ± 0.9	13.4 ± 1.9 clean	Wall remains low.

Table 10: Memory scaffold ablation. A six-step history probe is compared with a single-state probe on the existing sensor-agent hidden states.

Condition	Single	History	Gain	Takeaway
Normal sequence	0.662	0.619	-0.043	No clean gain.
Zero-frame break	0.537	0.667	+0.130	Partial recovery.
Freeze break	0.651	0.606	-0.045	No help for stale frames.
Occlusion-like gap	0.563	0.664	+0.102	Partly bridges gap.